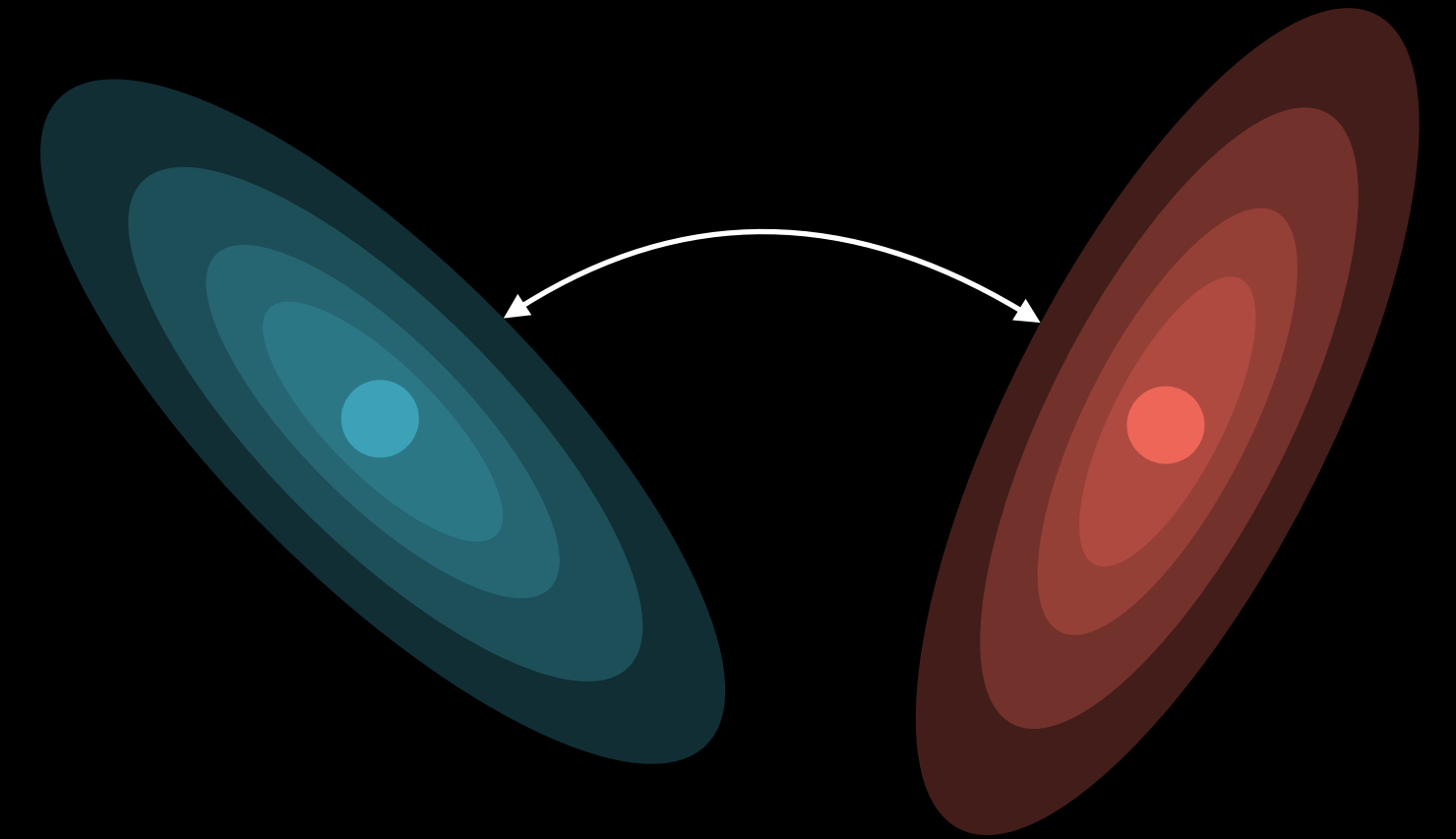


$$\alpha \equiv \mathcal{N}(\mu_{\alpha}, \Sigma_{\alpha})$$

$$\beta \equiv \mathcal{N}(\mu_\beta, \Sigma_\beta)$$

W_2 between ND Gaussians

$$\alpha = \mathcal{N}(\mu_\alpha, \Sigma_\alpha) \quad \beta = \mathcal{N}(\mu_\beta, \Sigma_\beta)$$



Closed form: $W_2^2(\alpha, \beta) = \|\mu_\alpha - \mu_\beta\|_2^2 + \mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2$

Where the second term is the *Bures* distance:

$$\mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2 = \text{tr}\left(\Sigma_\alpha + \Sigma_\beta - 2\left(\Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2}\right)^{1/2}\right)$$

If $\Sigma_\alpha, \Sigma_\beta$ commute, this further simplifies to:

$$\mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2 = \|\Sigma_\alpha^{1/2} - \Sigma_\beta^{1/2}\|_F^2$$

Gelbrich Bound on W_2

For **arbitrary** measures α and β with means/covars $\mu_\alpha, \Sigma_\alpha$ and μ_β, Σ_β